

William L. Couturiaux

Software Engineer III

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SUMMARY

Software engineer with an M.S. in Engineering Management and over 10 years of experience building desktop and web-based systems for JPMorgan Chase, General Motors, and Stellantis to automate complex engineering and operational workflows. Delivered enterprise tools scaled up to 300,000+ users, resolved multi-million-dollar plant-level risks, and drove 80%+ efficiency gains across data platforms.

SKILLS

React, Python, Angular, Java, Spring Boot, C#, .NET, TypeScript, AWS, Agentic Development, GitHub Copilot

WORK EXPERIENCE

Software Engineer III

June 2024–Present

JPMorgan (Asset and Wealth Management)

Columbus, OH

- Reduced manual production team workload by 15 hours per week by engineering Java/Spring Boot reconciliation workflows to automate position-break discovery across 71 accounts and 5,000 daily positions.
- Reduced transaction-replay time 5-10x by building a batch replay API to replace single transaction replay.
- Reduced UI load times from 7s to under 1s and eliminated RDS memory spikes by building a configuration-driven Java purge API to automate the cleanup of millions of records daily.
- Eliminated 2.5 hours of weekly administrative overhead for 60,000 managers by delivering a live Angular/Java dashboard that automated attendance tracking, task generation, and coaching actions.
- Achieved 100% pipeline state consistency and zero storage anomalies from launch by engineering a Python attachment service (AWS RDS, S3) featuring SHA-256 deduplication.

Senior Software Engineer (Engineering Tools)

June 2020–June 2024

General Motors

Milford, MI

- Saved \$28.8M in manufacturing downtime within 36 hours by implementing C# WPF dynamic data routing to securely isolate export-controlled records and unblock global issue-tracking.
- Saved 1,600 engineering hours by delivering .NET enhancement in 4 days after an upstream system migration introduced breaking data model changes.
- Reduced production validation lead time by 6 weeks per release cycle by re-architecting C# release logic to process asynchronous inputs, preventing downstream plant shutdowns.
- Increased active platform usage by over 160% (from 300 to 800+ weekly users) across 15 vehicle programs by developing a C#/.NET production readiness tool that prevented late-stage build conflicts.

Software Integration Engineer (Full-size Trucks and SUVs)

Nov 2018–June 2020

General Motors

Milford, MI

- Increased daily vehicle processing capacity by 200% and reduced per-vehicle scan time by 85% by engineering a Python ECU data extraction tool to eliminate peak release bottlenecks.
- Reduced vehicle data extraction time from 4 minutes to 28 seconds by engineering a Python/Selenium automation tool, bypassing API-less legacy systems to permanently eliminate manual lookups.
- Reduced a multi-day manual vehicle options re-configuration process to 3 minutes by engineering a VBA batch script to evaluate complex conditional option rules across directories of vehicle build sheets.
- Reduced release risk by ensuring software version consistency across distributed vehicle ECUs, identifying integration defects, and resolving CAN bus issues before functional safety validation.

Rotational Engineer Program and Controls & Calibration Engineer

June 2011–Nov 2018

Fiat Chrysler Automobiles (Stellantis)

Auburn Hills, MI

- Recovered \$300,000 in unallocated project capital by engineering a VBA automation tool to precisely reconcile cross-border transactions against historical transaction-date exchange rates.
- Improved vehicle driveability ratings by 0.5 points by building MATLAB tools to direct calibration workflows and automate the evaluation of telemetry data against industry best practices.
- Reduced mean time to repair (MTTR) by 45% by architecting the deployment infrastructure for iQagent, a mobile application replacing paper schematics with instant digital lookups.

PROJECTS

IRIS (International Resale Inventory System)

2025–Present

React, Node.js, PostgreSQL

- Engineered a cross-continental logistics application that eliminated 3 hours of manual data entry per shipment, replacing legacy spreadsheets with automated WooCommerce synchronization to reconcile international shipments and guarantee real-time inventory accuracy.
- Eliminated over 200 hours of manual copywriting by engineering a Node.js product creation pipeline that integrates Gemini LLM to dynamically generate descriptions for a 600+ item catalog.

NASCAR Driver Performance & Analytics Dashboard - GM Hackathon

2023

React, Java, Spring Boot

- Co-engineered a real-time NASCAR driver telemetry platform formally adopted by GM Racing, winning 1st place in a GM Hackathon by synchronizing high-frequency time-series data with spatial track coordinates using React and Spring Boot.

LEADERSHIP & ACHIEVEMENTS

- Mentored students through product ideation and prototyping of inclusive mobile applications, focusing on real-world problem framing and building technical confidence. (Black Girls Code - 2020)

EDUCATION

Oakland University - Rochester, MI

- Master of Science in Engineering Management

University of Pittsburgh - Pittsburgh, PA

- Bachelor of Science in Industrial Engineering